

On Modal Logic of Full Products of Neighborhoods Frames

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3. Key ideas sketched 15 Minutes

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Motivation/Main results 4 Minutes

Introduction

The CleanEasy Beamer theme provides a clean, professional look for academic and business presentations.

Key features:

- Clean, minimalist design
- Carefully chosen color scheme
- Professional typography
- Flexible block environments
- Customizable footers and section pages

About this template

This presentation serves as both documentation and demonstration, showing the various elements and features available in the CleanEasy theme.

Block environments

Example block

Example blocks can be used to showcase code, examples, or case studies.

Alert block

Alert blocks draw attention to critical information, warnings, or caveats.

Theorem

In a right-angled triangle, the square of the hypotenuse equals the sum of squares of the other two sides.

Definition

A prime number is a natural number greater than 1 that cannot be formed by multiplying two smaller natural numbers.

Preliminaries 5 Minutes

Neighbourhood frame

Definition

Let X be a non-empty set and $k \geq 1$. A k -neighborhood frame (k -n-frame) is a pair $(X, \vec{\tau})$, where $\vec{\tau} = (\tau_1, \dots, \tau_k)$ and each $\tau_i : X \rightarrow 2^{2^X}$ assigns to every $x \in X$ a *filter* $\tau_i(x)$ on X , i.e.

- $\tau(x) \neq \emptyset$
- if $U \in \tau(x)$ and $U \subseteq V$ then $V \in \tau(x)$
- if $U, V \in \tau(x)$ then $U \cup V \in \tau(x)$

Full Product of Frames

Definition

Let $\mathcal{X} = (X, \tau_1)$ and $\mathcal{Y} = (Y, \tau_2)$ be two unimodal neighborhood frames. Then the full product of these two frames is a 3-n-frame $\mathcal{X} \times_n \mathcal{Y} = (\mathbf{X} \times \mathbf{Y}, \tau'_1, \tau'_2, \tau)$ defined as follows :

$$\tau'_1(x, y) = \{U \subseteq \mathbf{X} \times \mathbf{Y} \mid \exists V \in \tau_1(x) : V \times \{y\} \subseteq U\},$$

$$\tau'_2(x, y) = \{U \subseteq \mathbf{X} \times \mathbf{Y} \mid \exists V \in \tau_2(y) : \{x\} \times V \subseteq U\},$$

$$\tau(x, y) = \{U \subseteq \mathbf{X} \times \mathbf{Y} \mid \exists V \in \tau_1(x) \exists W \in \tau_2(y) : V \times W \subseteq U\}$$

Bounded morphism

Definition

Let $F = (W, \tau_1, \dots, \tau_k)$ and $F' = (W', \sigma_1, \dots, \sigma_k)$ be neighborhood frames. A function $f : W \rightarrow W'$ is called a bounded morphism if

- for all $1 \leq i \leq k$, for all $w \in W$ and all $U \in \tau_i(w)$, there is a $V \in \sigma_i(f(w))$ with $V = f(U)$,
- for all $1 \leq i \leq k$, for all $w \in W$ and all $V \in \sigma_i(f(w))$, there exists $U \in \tau_i(w)$ such that $f(U) \subseteq V$.

Note: If $F \twoheadrightarrow F'$, then $\text{Log}(\{F\}) \subseteq \text{Log}(\{F'\})$.

Definition

Let L_1 and L_2 be two unimodal logics. Then the *fusion* ($\otimes$) of these logics is

$$K_2 + (L_1)_{\Box \rightarrow \Box_1} + (L_2)_{\Box \rightarrow \Box_2}$$

Furthermore the full product ($\times_n^+$) is defined as

$$L_1 \times_n^+ L_2 = \text{Log}\{\mathcal{X} \times \mathcal{Y} \mid \mathcal{X} \models L_1 \text{ and } \mathcal{Y} \models L_2\}$$

Lists and Numbering

Bulleted list:

- First level item
 - Second level item
 - Another second level
 - Third level item
- Another first level item

Numbered list:

1. First step
 - 1.1 Substep one
 - 1.2 Substep two
2. Second step
3. Third step

Tables

Table: Sample table with booktabs style

Header 1	Header 2	Header 3
Row 1, Col 1	Row 1, Col 2	123.45
Row 2, Col 1	Row 2, Col 2	67.89
Row 3, Col 1	Row 3, Col 2	456.78

Table styling

The CleanEasy theme works well with the booktabs package for professional-looking tables. Simple color alterations make tables more readable without being distracting.

Mathematical Equations

The CleanEasy theme includes proper mathematical typesetting:

$$E = mc^2 \tag{1}$$

$$F = G \frac{m_1 m_2}{r^2} \tag{2}$$

Maxwell's equations in differential form:

$$\nabla \cdot \vec{E} = \frac{\rho}{\varepsilon_0} \tag{3}$$

$$\nabla \cdot \vec{B} = 0 \tag{4}$$

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \tag{5}$$

$$\nabla \times \vec{B} = \mu_0 \vec{J} + \mu_0 \varepsilon_0 \frac{\partial \vec{E}}{\partial t} \tag{6}$$

Inline equations like $E = mc^2$ are also properly rendered.

Code Listings

```
1 # A simple Python function
2 def fibonacci(n):
3     """Return the nth Fibonacci number"""
4     if n <= 0:
5         return 0
6     elif n == 1:
7         return 1
8     else:
9         a, b = 0, 1
10        for _ in range(2, n + 1):
11            a, b = b, a + b
12        return b
13
14 # Calculate the 10th Fibonacci number
15 result = fibonacci(10)
16 print(f"The 10th Fibonacci number is {result}")
```

Figures and Graphics



Figure: Sample placeholder image

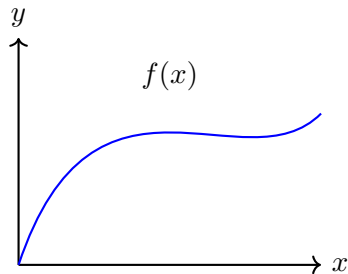


Figure: Simple TikZ diagram

Overlays and Animations

Beamer supports step-by-step revelations:

- First point appears on slide 1

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Beamer supports step-by-step revelations:

- First point appears on slide 1
- Second point appears on slide 2

This text appears after a pause.

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This text appears after a pause.

And this content appears on slide 4.

Overlays and Animations

Beamer supports step-by-step revelations:

- First point appears on slide 1
- Second point appears on slide 2
- Third point appears on slide 3

This text appears after a pause.

And this content appears on slide 4.

Delayed Block

This entire block appears only on slide 5.

Citations and References

CleanEasy works well with bibliographies and citations:

Sample citation

According to Einstein [1], space and time are relative.

Bibliography management

The theme is compatible with BibTeX, BibLaTeX, and other bibliography management tools.

Custom TikZ Graphics

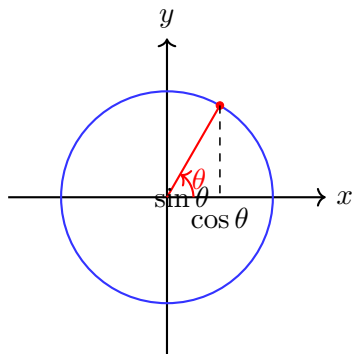


Figure: The unit circle with trigonometric functions

Key ideas sketched 15 Minutes

Main Theorem

Theorem

Let Λ be a unimodal logic with $\Lambda \in \{T, D\}$. Then the following holds:

$$\Lambda \otimes \Lambda \otimes \Lambda + \Box p \rightarrow \Box_1 \Box_2 p \wedge \Box_2 \Box_1 p = \Lambda \times_n^+ \Lambda$$

In the following we will sketch the proof for $\Lambda = T$.

We denote $T \otimes T \otimes T + \Box p \rightarrow \Box_1 \Box_2 p \wedge \Box_2 \Box_1 p$ as T_3m

Proof steps

- Idea: create an infinite branching and depth Kripke tree with three relations (called $T_{\omega,\omega,\omega}^{r,m}$) with $\text{Log}(T_{\omega,\omega,\omega}^{r,m}) = T_3m$

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- Note: For any Kripke frame $F = (W, R_1, \dots, R_n)$, we can construct a neighborhood frame F' , i.e. for any $w \in W$: $\tau(w) = \{U \mid R_1(w) \subseteq U \subseteq W\}$ where $Log(F) = Log(F')$
- construct neighbourhood product $N_{\omega}^r \times_n^+ N_{\omega}^r$ s.t. there is surjective bounded morphism from $N_{\omega}^r \times_n^+ N_{\omega}^r$ to $N(T_{\omega,\omega,\omega}^{r,m})$

$$T \otimes T \otimes T + (\Box p \rightarrow \Box_1 \Box_2 p \wedge \Box_2 \Box_1 p) = T_3m$$

$$\text{Log}(T_{\omega,\omega,\omega}^{r,m}) = T_3 m$$

Theme Customization

The CleanEasy theme can be easily customized:

- Edit `beamercolorthemeCleanEasy.sty` to change colors
- Modify `beamerfontthemeCleanEasy.sty` for different fonts
- Adjust `beamerinnerthemeCleanEasy.sty` for layout changes
- Update `configs.tex` for footer and section page customization

Important Note

Always maintain consistent design elements throughout your presentation for a professional look.

Final Thoughts

Benefits of CleanEasy

- Professional appearance suitable for academic and business contexts
- Careful attention to typography and spacing
- High readability with suitable contrast ratios
- Flexible design that works with different content types

The CleanEasy theme is designed to let your content shine without distractions

Thank you!

`your@email.com`

`https://someurl.com`

References



Albert Einstein. *On the Electrodynamics of Moving Bodies*. Annalen der Physik, 1905.



Till Tantau. *The Beamer Class*. <https://ctan.org/pkg/beamer>

Open questions 1 Minute
